

DBMS LAB – SQL TRANSACTIONS (2)

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```
mysql> -- Q1
mysql> INSERT INTO employees
    -> VALUES (108,'Arjun',1,101,52000);
ERROR 1062 (23000): Duplicate entry '108' for key 'employees.PRIMARY'
mysql>
mysql> SELECT * FROM employees;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | manager_id | salary |
+-----+-----+-----+-----+-----+
| 101 | Rahul | 1 | NULL | 96800.00 |
| 102 | Anita | 1 | 101 | 72600.00 |
| 103 | Vijay | 2 | 101 | 55000.00 |
| 104 | Sneha | 2 | 103 | 45000.00 |
| 105 | Kiran | 3 | 101 | 70000.00 |
| 106 | Meena | 1 | 102 | 78650.00 |
| 107 | Arjun | 1 | 101 | 62920.00 |
| 108 | Arjun | 1 | 101 | 57200.00 |
+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql>
mysql>
mysql> -- Q2
mysql> UPDATE employees
    -> SET salary = salary * 1.10
    -> WHERE dept_id = (
    ->     SELECT dept_id
    ->     FROM departments
    ->     WHERE dept_name = 'CSE'
    -> );
Query OK, 5 rows affected (0.01 sec)
Rows matched: 5  Changed: 5  Warnings: 0
```

```
mysql> DELETE FROM employees
-> WHERE salary < 30000;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM employees;
```

emp_id	emp_name	dept_id	manager_id	salary
101	Rahul	1	NULL	106480.00
102	Anita	1	101	79860.00
103	Vijay	2	101	55000.00
104	Sneha	2	103	45000.00
105	Kiran	3	101	70000.00
106	Meena	1	102	86515.00
107	Arjun	1	101	69212.00
108	Arjun	1	101	62920.00

8 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> -- Q4
mysql> SELECT * FROM employees;
```

emp_id	emp_name	dept_id	manager_id	salary
101	Rahul	1	NULL	106480.00
102	Anita	1	101	79860.00
103	Vijay	2	101	55000.00
104	Sneha	2	103	45000.00
105	Kiran	3	101	70000.00
106	Meena	1	102	86515.00
107	Arjun	1	101	69212.00
108	Arjun	1	101	62920.00

8 rows in set (0.00 sec)

```
-> FROM departments d
-> LEFT JOIN employees e
-> ON d.dept_id = e.dept_id;
```

dept_name	emp_name
CSE	Arjun
CSE	Arjun
CSE	Meena
CSE	Anita
CSE	Rahul
ECE	Sneha
ECE	Vijay
MECH	Kiran
Civil	NULL

9 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> -- Q8
mysql> SELECT e.emp_name AS Employee,
-> m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id = m.emp_id;
```

Employee	Manager
Rahul	NULL
Anita	Rahul
Vijay	Rahul
Sneha	Vijay
Kiran	Rahul
Meena	Anita
Arjun	Rahul
Arjun	Rahul

8 rows in set (0.00 sec)

```
mysql> -- Q10
mysql> SELECT dept_name,
->         AVG(salary) AS AverageSalary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY dept_name;
```

dept_name	AverageSalary
CSE	80997.400000
ECE	50000.000000
MECH	70000.000000

3 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> -- Q11
mysql> SELECT dept_name,
->         AVG(salary) AS AverageSalary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY dept_name
-> HAVING AVG(salary) > 60000;
```

dept_name	AverageSalary
CSE	80997.400000
MECH	70000.000000

2 rows in set (0.00 sec)

```
mysql>
mysql>
mysql> -- Q12
mysql> SELECT MAX(salary) AS HighestSalary,
->         MIN(salary) AS LowestSalary
-> FROM employees;
```

HighestSalary	LowestSalary
106480.00	45000.00

1 row in set (0.00 sec)

```

mysql> -- Q16
mysql> SELECT dept_name,
->         COUNT(*) AS EmployeeCount
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY dept_name
-> HAVING COUNT(*) > 5;
Empty set (0.00 sec)

mysql>
mysql>
mysql> -- Q17
mysql> SELECT emp_name, salary
-> FROM (
->     SELECT emp_name,
->            salary,
->            DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
->     FROM employees
-> ) AS ranked
-> WHERE rnk = 2;
+-----+-----+
| emp_name | salary |
+-----+-----+
| Meena    | 86515.00 |
+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> -- Q18
mysql> SELECT emp_name, salary
-> FROM employees e
-> WHERE salary > (
->     SELECT AVG(salary)
->     FROM employees
->     WHERE dept_id = e.dept_id
-> );
+-----+-----+
| emp_name | salary |
+-----+-----+
| Rahul    | 106480.00 |
| Vijay    | 55000.00 |
| Meena    | 86515.00 |
+-----+-----+
3 rows in set (0.00 sec)

```